# Integrated molecular, structural, and functional brain atlases in the NiSpace toolbox

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Application? Poster **2811**

Brain & Behaviour

## NiSpace

- Multimodal **reference brain atlases** are widely used to **contextualize neuroimaging** findings,[1,2] but are scattered across **heterogeneous** sources, formats, imaging spaces, and parcellations, limiting **reproducibility**.[3]
- NiSpace (**NeuroImaging Spatial Colocalization Environment**) is an open-source **Python toolbox** for **spatial inference on brain maps**.[4]
- We introduce its **curated, standardized library** of multimodal brain datasets and parcellations.
- Beyond dataset access, NiSpace provides user-friendly **end-to-end workflows** for spatial **correlation**, **significance** testing (spatial surrogates, group, set permutation), **imaging space transforms**, and **visualization**.

### Getting started

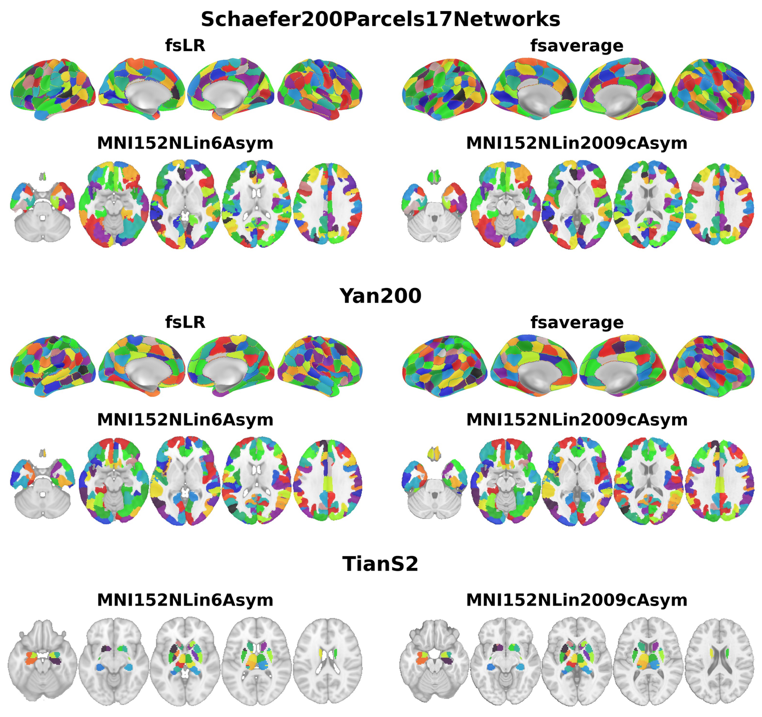
```
pip install git+https://github.com/LeonDLotter/NiSpace.git@dev

from nispace.workflows import colocalization, xsea

# Colocalize with selected nuclear imaging maps
nsp = colocalization(y="your_map.nii.gz", x="pet")
pvalues = nsp.get_p_values()

# X set enrichment analysis (XSEA) with cell markers
nsp = xsea(y="your_map.nii.gz", x="magic")
```

## Parcellations

- NiSpace supports **four standard imaging spaces** (MNI152NLin6Asym, MNI152NLin2009cAsym, fsLR, fsaverage) and currently includes **27 cortical and subcortical parcellations**. New parcellations are continuously added.
- All parcellations are **available in all four spaces** (subcortical: MNI only).
- **Cortex and subcortex** (excl. cerebellum) are **managed separately**, supporting flexible combination of parcellated datasets and multi-method spatial surrogate generation.
- Homogenized region labelling allows for operations across all parcellations in all spaces.

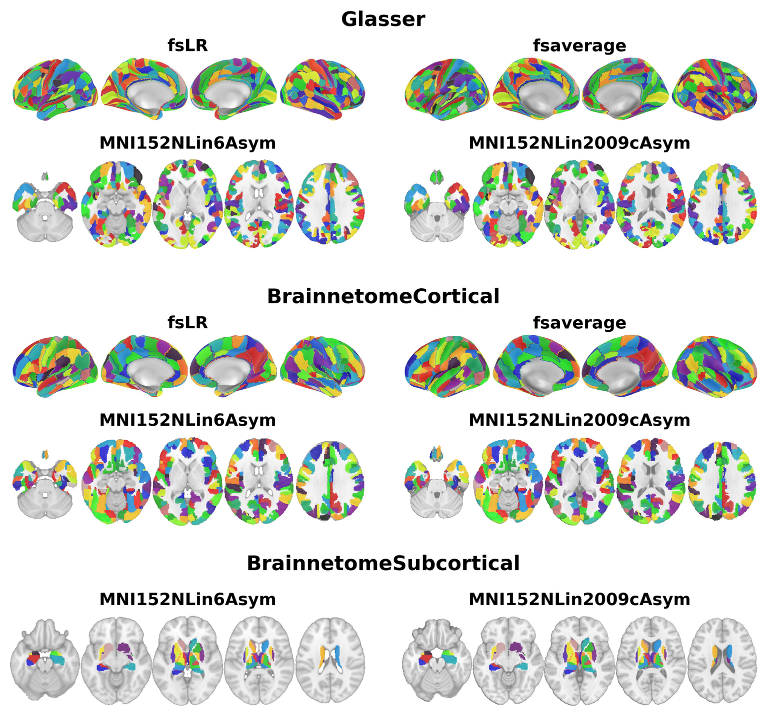
### Functional

Schaefer/Yan: 100-1000, Tian: 16-54 parcels.



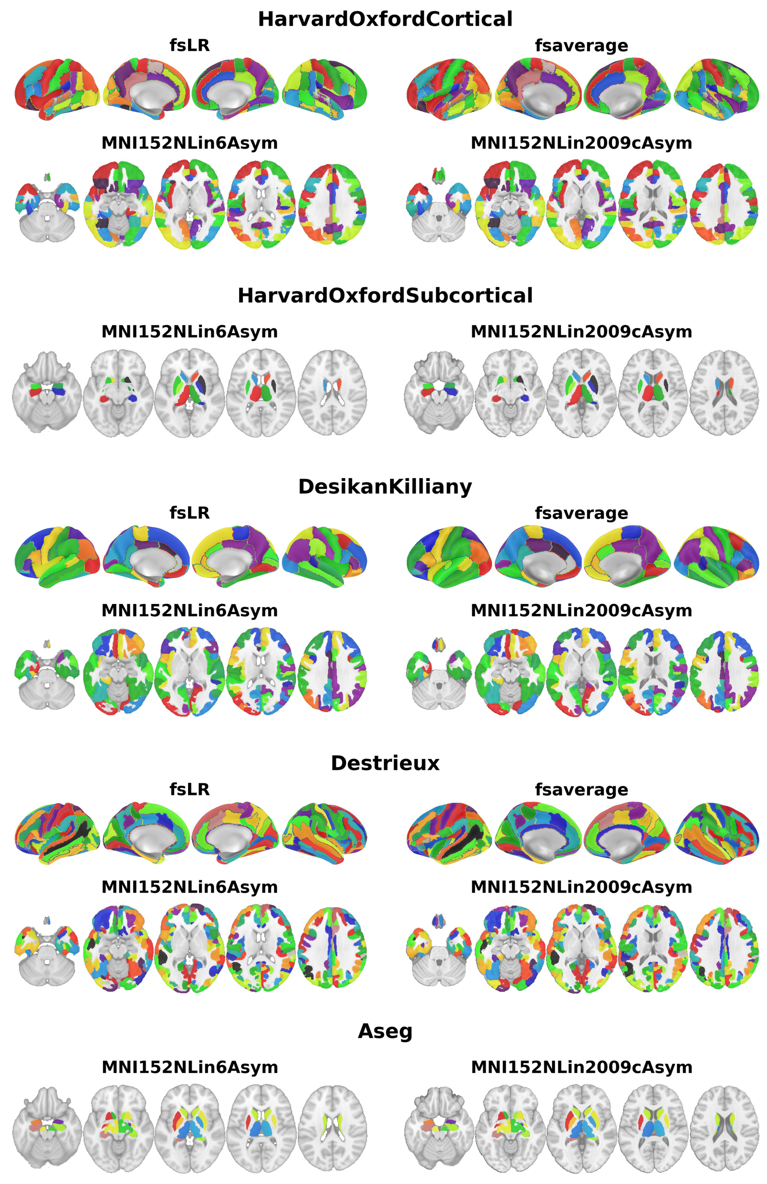

### High-density anatomical/multimodal

Glasser: 360, Brainnetome: 210+36 parcels.




### Macroscale anatomical

HO: 96+12, DK: 68, Destrieux: 148, Aseg: 16 parcels.




## Reference atlases

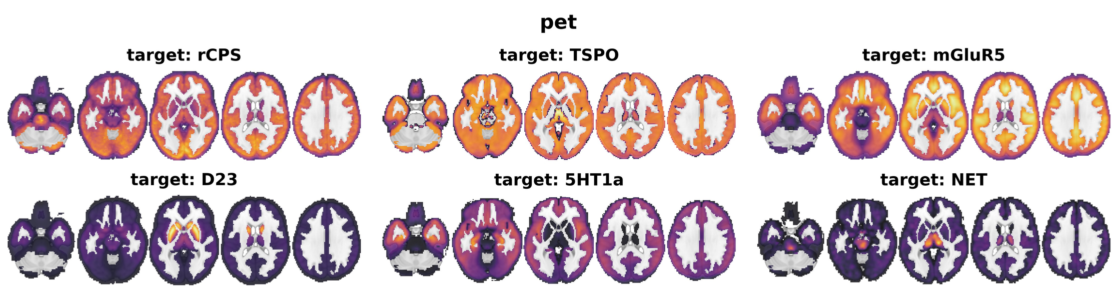
- Reference datasets can be:
  - **Individual maps** in either all four, or only surface spaces.
  - **Parcellated tables** for all available parcellations (derived from maps), or only for a specific source parcellation.
- Reference datasets are accompanied by **collections**:
  - **Subsets of maps** within a reference dataset (e.g., one map per PET target, selected Neurosynth terms, …).
  - **Curated sets of maps** for averaging and set enrichment analysis (e.g., nuclear imaging maps grouped by target, Neurosynth maps grouped by cognitive domain, …).

### Missing something?

If you would like to have a specific dataset or parcellation added to NiSpace, open a GitHub issue – we will evaluate and integrate it.

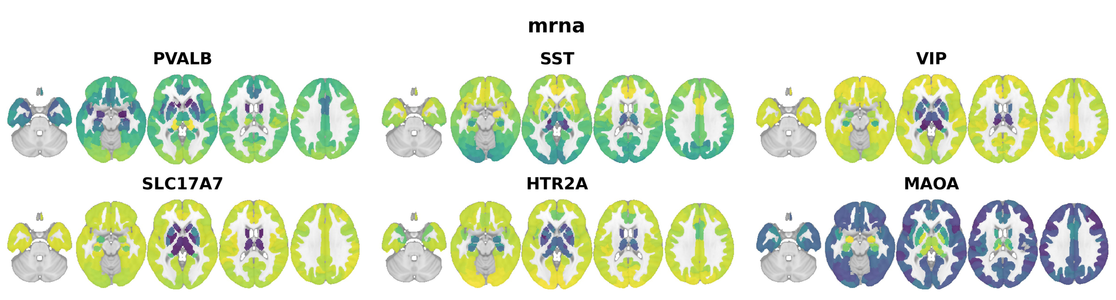
### Molecular brain atlases

"pet": **49 nuclear imaging maps**, 29 molecular targets, 37 tracers. **Nonlinearly co-registered** to specific MNI and surface templates. **Collections** by tracer, by target, by family; curated subsets.

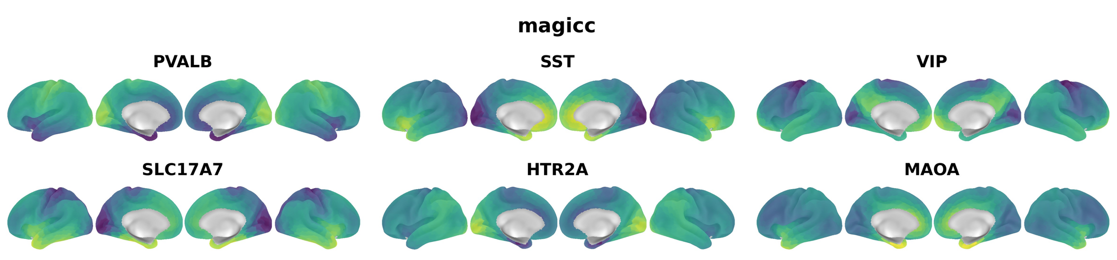



*(fsLR, fsaverage, MNI152NLin6Asym, MNI152NLin2009cAsym, all parcellations)*

"mrna"|"magicc": >11k gene expression maps, reconstructed on **parcel-level ("mrna")** or continuously on the **fsLR surface ("magic")**. **Collections** by cell type markers, GeneOntology, SynGO, BrainSpan, …



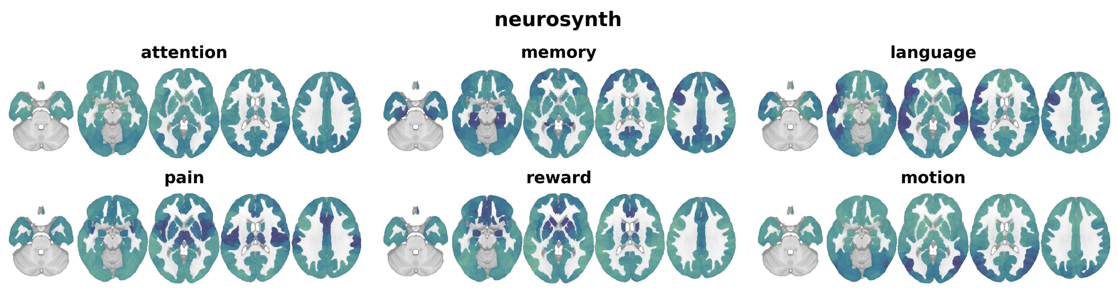

*(all parcellations)*




*(all cortex parcellations)*

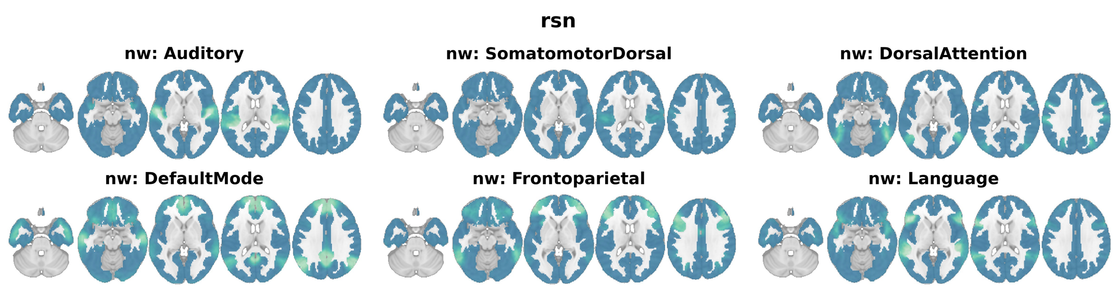
### Functional brain atlases

"neurosynth": >1k curated **meta-analytic** Neurosynth term maps. **Collection** of selected terms sorted by cognitive domains.

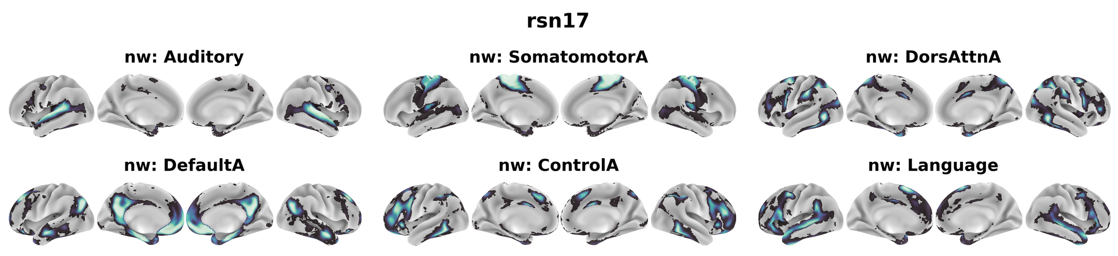



*(all parcellations)*

"rsn"|"rsn17": **resting-state network probability** maps in MNI ("rsn", n=14, Dworetsky) or surface ("rsn17", n=17, Kong).



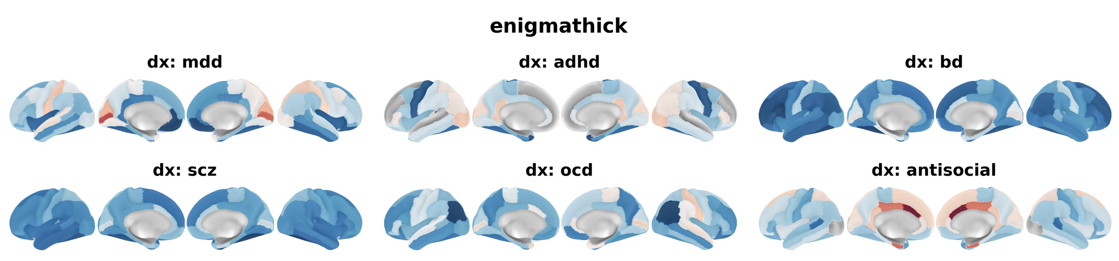

*(fsLR, fsaverage, MNI152NLin6Asym, MNI152NLin2009cAsym, all cortex parcellations)*



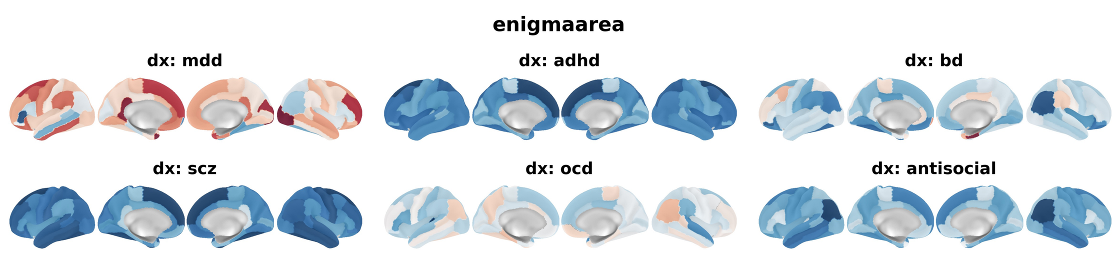

*(fsLR, fsaverage, all cortex parcellations)*

### Brain alterations

"enigmathick"|"enigmaarea": **ENIGMA** cortical thickness, surface area, and subcortical volume **effect size** maps. **Collection** of the main comparison in each source study.




*(Aseg, DesikanKilliany)*



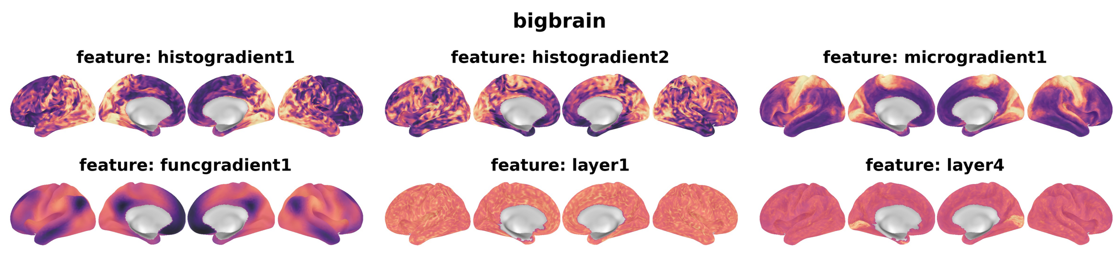

*(Aseg, DesikanKilliany)*

### Fetching data

```
from nispace import datasets

# Fetch a parcellation (returns a custom Parcellation class)
parc = datasets.fetch_parcellation("Glasser")
nifti = parc.get_image(space="MNI152NLin6Asym")

# Fetch a reference dataset (returns file lists or DataFrames)
list_of_giftis = datasets.fetch_reference("pet", space="fsLR")
table = datasets.fetch_reference("pet", parcellation="Glasser")
```

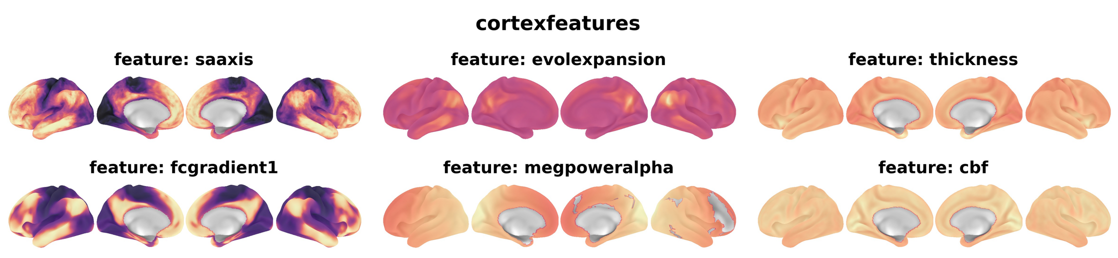
### Multimodal brain organization

"bigbrain": 13 postmortem & in-vivo maps of **cortex differentiation**. **Collections** for differentiation gradient and layer subsets.



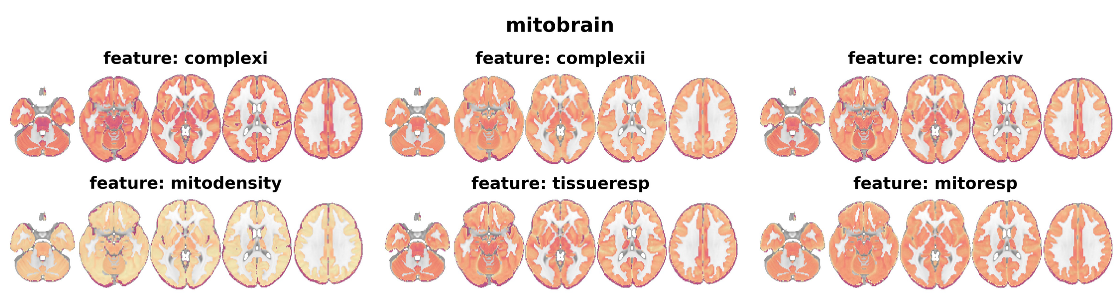

*(fsLR, fsaverage, all cortex parcellations)*

"cortexfeatures": 23 multimodal **MRI, MEG, & PET** maps. **Collections** for cortical organization, metabolism, MEG power.



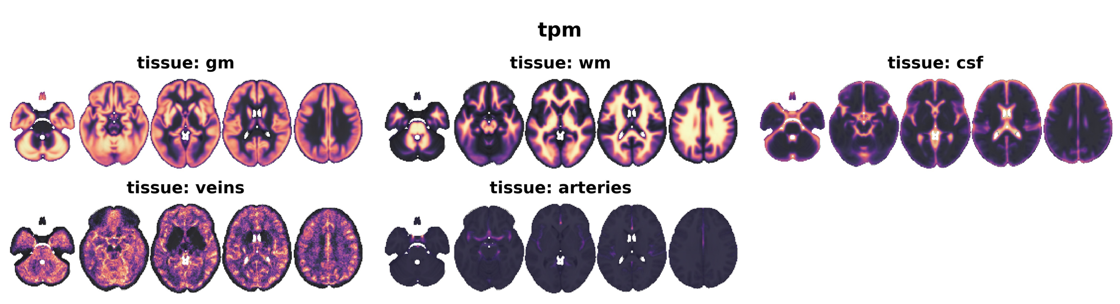

*(fsLR, fsaverage, all cortex parcellations)*

"mitobrain": 6 postmortem maps of **mitochondrial features**.




*(fsLR, fsaverage, MNI152NLin6Asym, MNI152NLin2009cAsym, all parcellations)*

"tpm": 5 tissue probability maps used as **brain-level covariates**.




*(fsLR, fsaverage, MNI152NLin6Asym, MNI152NLin2009cAsym, all parcellations)*

## Licensing and attribution

- Included datasets carry a range of **open licenses**. To accommodate these, the NiSpace-data repository is distributed under the restrictive **CC BY-NC-SA 4.0**.
- To **support correct attribution**, every API call to retrieve references or parcellations automatically prints a **dataset description** and **primary citations with DOIs**, as well as a license and **DOI for each individual map**.

## Reproducibility and versioning

- NiSpace data are stored in a **separate public GitHub repository**: LeonDLotter/NiSpace-data.
- NiSpace pins the **data repository commit hash** and holds a **library of file hashes** associated with that commit.
- Files are downloaded lazily, **cached locally**, and resolve against the fixed data hashes; mismatches force re-download: every **NiSpace** version uses **a fixed dataset**.

## Resources

| NiSpace | Poster PDF | All citations |
|---|---|---|

**Text references:** (1) Misic B, *Apert Neuro*, 2025. | (2) Markello RD et al., *Nat Methods, 2022* | (3) Royer J et al., *Nat Rev Neurosci*, 2026. | (4) Lotter LD & Dukart J, *Zenodo*, 2024. | **Parcellation and dataset references:** See QR-code.
**Acknowledgments**: LDL is supported by BMBF and MPG, Germany. LDL, RB, SBE and JD receive support from the German Research Foundation (DFG; DU 2027/7-1 project number 549186835).